FPGA_power

File: FPGA_Power.kicad_sch

CMOS_power

File: power.kicad_sch

current_generators

File: current_generators.kicad_sch

MarsAX3_SoM_IO

File: MarsAX3_SoM_IO.kicad_sch

DDR4_Edge_Connector

File: DDR4_Edge_Connector.kicad_sch

eye_tech_sensor

File: eye_tech_sensor.kicad_sch

acquisition

File: acquisition.kicad_sch

MarsAX3_SoM_Pwr_Cfg

File: MarsAX3_SoM_Pwr_Cfg.kicad_sch

P/N: e10009-002

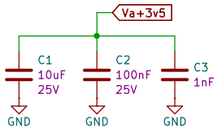
Elios S.r.l.

Sheet: /
File: e10009-002-00A.kicad_sch

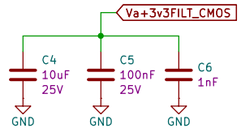
Title: E2D-GS Interface

Size: A3	Date: 2025-05-21	**Rev: 00A**
KiCad E.D.A. 9.0.1		Id: 1/9

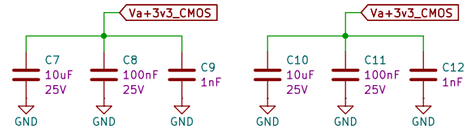
These caps go as close as possible to VDDH pin of the EyeTech CMOS



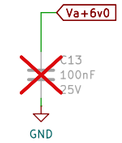
These caps go as close as possible to VDD pin of the EyeTech CMOS



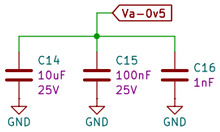
These caps go as close as possible to VDDOx pin of the EyeTech CMOS



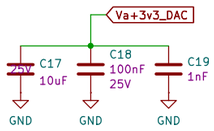
Place capacitor near ADA4522



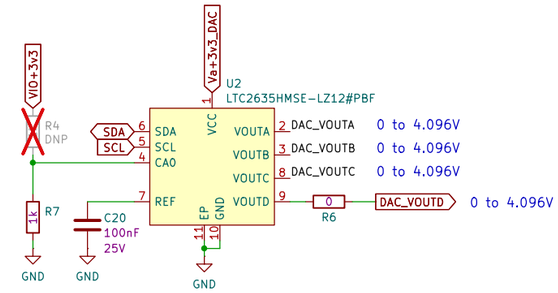
These caps go as close as possible to VSSH pin of the EyeTech CMOS



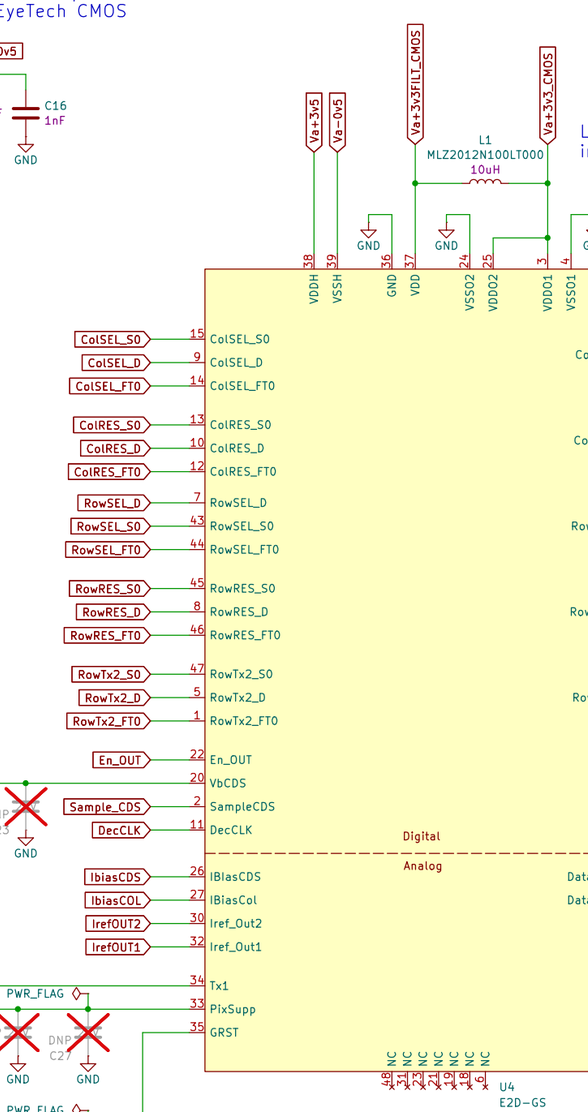
Place capacitors as close as possible to DAC



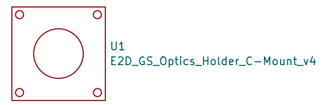
DAC for VbCDS, PixSupp supply and GRST

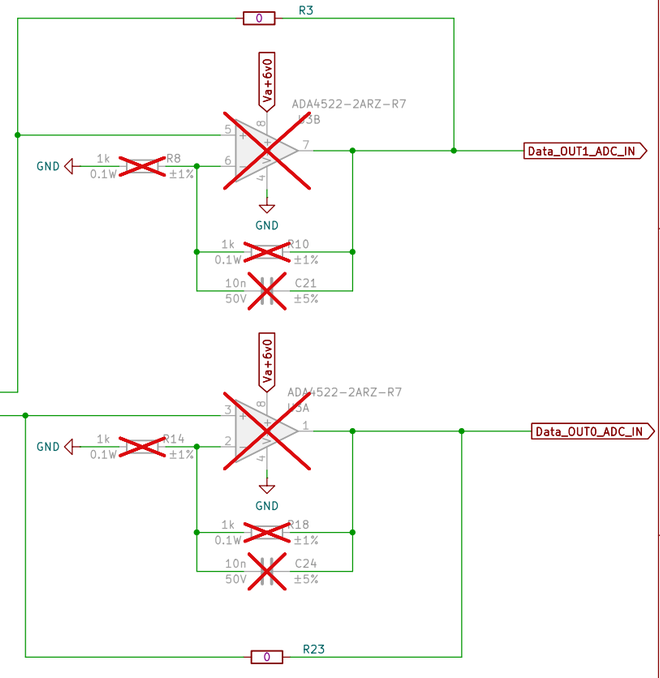


L1 with decoupling capacitors implements a noise filter

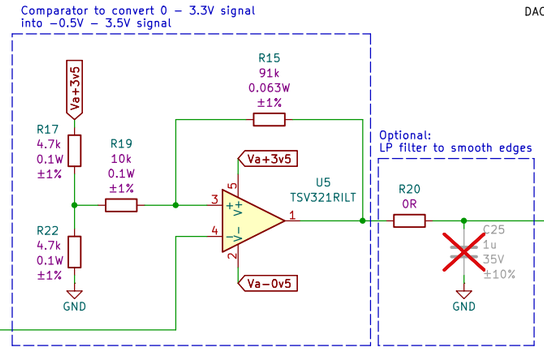


Optic holder

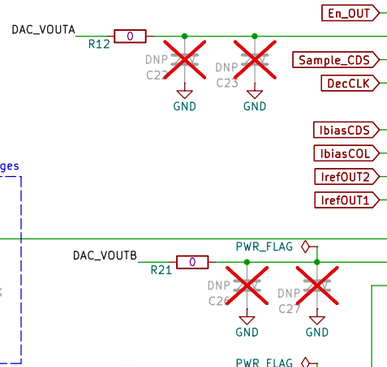




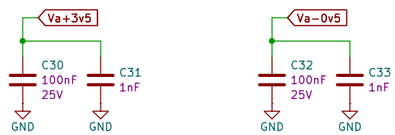
Comparator to convert 0 – 3.3V signal into −0.5V – 3.5V signal



Optional: LP filter to smooth edges



These caps go as close as possible to TSV321



Va+3v3FILT_CMOS PWR_FLAG

P/N: e10009-002

Elios S.r.l.

Sheet: /eye_tech_sensor/
File: eye_tech_sensor.kicad_sch

Title: E2D-GS Interface

Size: A3	Date: 2025-05-21	Rev: 00A
KiCad E.D.A. 9.0.1		Id: 2/9

Place at least the 100nF and 1nF right
at the supply pin of the respective amplifiers

Va+3v_ADC
C34 10uF 25V
C35 100nF
C36 1nF
GND

LTC6253HMS8#PBF
Va+3v_ADC
Data_OUT0_ADC_IN
R31 0R
GND
U6A
R29 22 0.1W ±1%
C52 0.33u 50V ±10%
R28 1k 0.063W ±1%
R30 22 0.1W ±1%
R32 49.9 0.063W ±1%
R33 1k 0.063W ±1%
C53 15p 50V ±2%
R34 0R
100V ±10% C54 1.5n
R36 1k 0.063W ±1%
R37 22 0.1W ±1%
C55 100n 25V ±10%
C56 10u 10V ±20%
R38 1k 0.063W ±1%
C57 15p 50V ±2%

C43 10uF 25V
C44 100nF
C45 100nF
C46 100nF
C47 10uF 25V
C51 100nF
R25
R26
R27 0

U7
AD9235BRUZ-65
OTR MODE SENSE VREF REFB REFT AVDD AGND VIN+ VIN− AGND AVDD CLK PDWN
D11 D10 D9 D8 DRVDD DGND D7 D6 D5 D4 D3 D2 D1 D0
ADC1_D11 ADC1_D10 ADC1_D9 ADC1_D8 ADC1_D7 ADC1_D6 ADC1_D5 ADC1_D4 ADC1_D3 ADC1_D2 ADC1_D1 ADC1_D0
VIO+3v3
ADC1_D[0..11]
R35 1k
PWDN_ADC
ADC_CLK

Decoupling for ADCs, place close to each power pin
VIO+3v3
C37 10uF 25V
C38 100nF
C39 1nF
Va+3v_ADC
C40 10uF 25V
C41 100nF
C42 1nF
C48 10uF 25V
C49 100nF
C50 1nF

R39 10k

LTC6253HMS8#PBF
Data_OUT1_ADC_IN
R46 0R
R44 22 0.1W ±1%
C70 0.33u 50V ±10%
R43 1k 0.063W ±1%
R45 22 0.1W ±1%
R47 49.9 0.063W ±1%
R48 1k 0.063W ±1%
C74 15p 50V ±2%
R49 0R
100V ±10% C75 1.5n
R51 1k 0.063W ±1%
R52 22 0.1W ±1%
C76 100n 25V ±10%
C77 10u 10V ±20%
R53 1k 0.063W ±1%
C78 15p 50V ±2%

C58 10uF 25V
C59 100nF
C66 100nF
C67 100nF
C68 10uF 25V
C69 100nF
R40
R41
R42 0

U8
AD9235BRUZ-65
OTR MODE SENSE VREF REFB REFT AVDD AGND VIN+ VIN− AGND AVDD CLK PDWN
D11 D10 D9 D8 DRVDD DGND D7 D6 D5 D4 D3 D2 D1 D0
ADC2_D11 ADC2_D10 ADC2_D9 ADC2_D8 ADC2_D7 ADC2_D6 ADC2_D5 ADC2_D4 ADC2_D3 ADC2_D2 ADC2_D1 ADC2_D0
VIO+3v3
ADC2_D[0..11]
R50 1k
PWDN_ADC
ADC_CLK

Decoupling for ADCs, place close to each power pin
VIO+3v3
C60 10uF 25V
C61 100nF
C62 1nF
Va+3v_ADC
C63 10uF 25V
C64 100nF
C65 1nF
C71 10uF 25V
C72 100nF
C73 1nF

Table 7. Reference Configuration Summary

Selected Mode	SENSE Voltage	Internal Switch Position	Resulting VREF (V)	Resulting Differential Span (V p-p)
External Reference	AVDD	N/A	N/A	2 × External Reference
Internal Fixed Reference	VREF	SENSE	0.5	1.0
Programmable Reference	0.2 V to VREF	SENSE	0.5 × (1 + R2/R1)	2 × VREF (See Figure 40)
Internal Fixed Reference	AGND to 0.2 V	Internal Divider	1.0	2.0

P/N: e10009-002

Elios S.r.l.

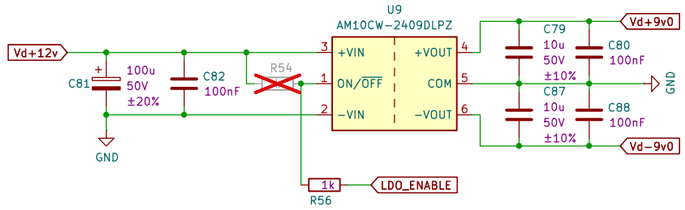
Sheet: /acquisition/
File: acquisition.kicad_sch

Title: E2D-GS Interface

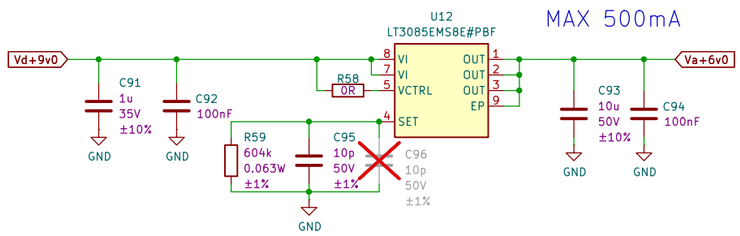
Size: A3 | Date: 2025-05-21 | Rev: 00A
KiCad E.D.A. 9.0.1 | Id: 3/9

CMOS Sensor Power supply

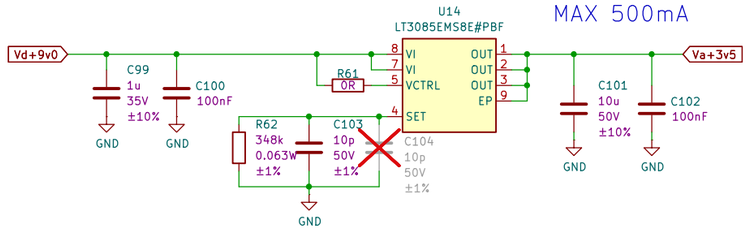
MAX 550mA each side



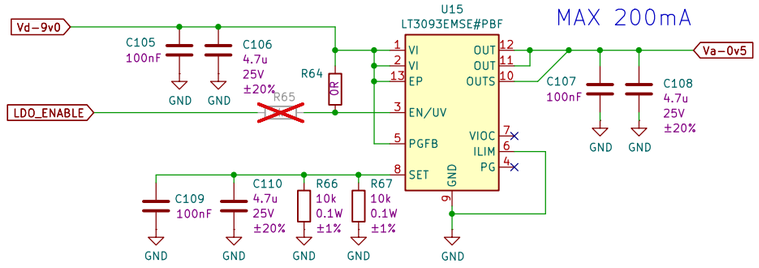
MAX 500mA



MAX 500mA

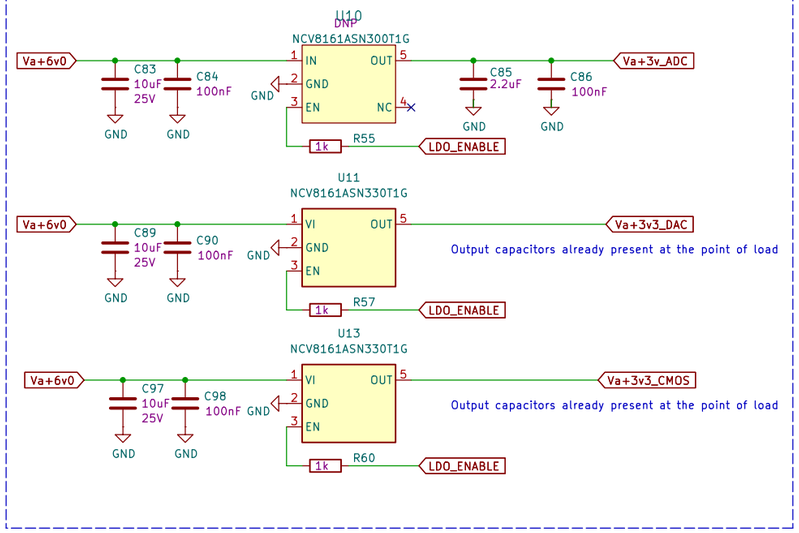


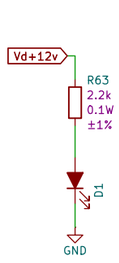
MAX 200mA



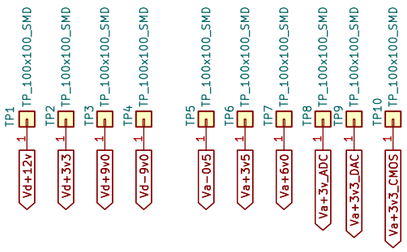
Place these close to the respective acquisition circuit

MAX 450mA each

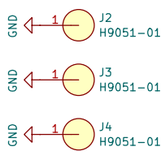


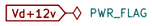


Power Supply Test Point



GND Turrets





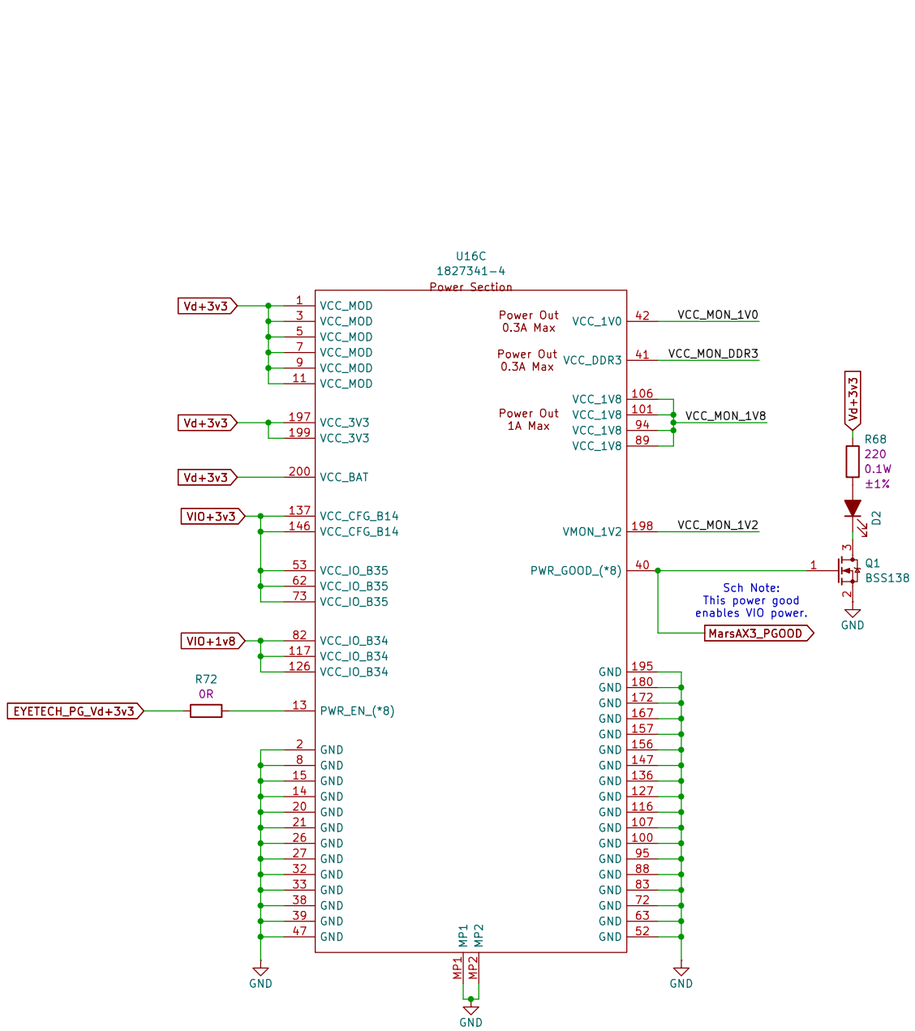
P/N: e10009-002

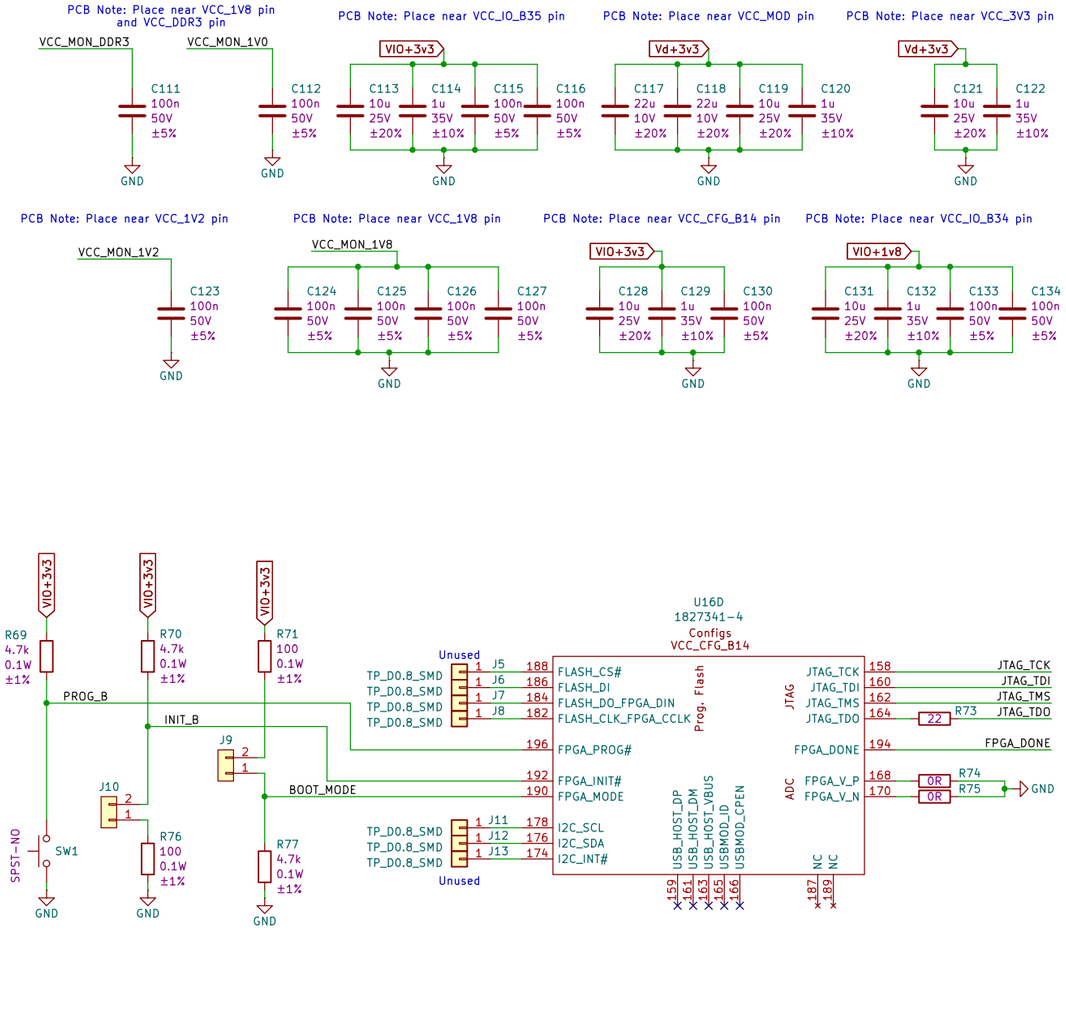
Elios S.r.l.

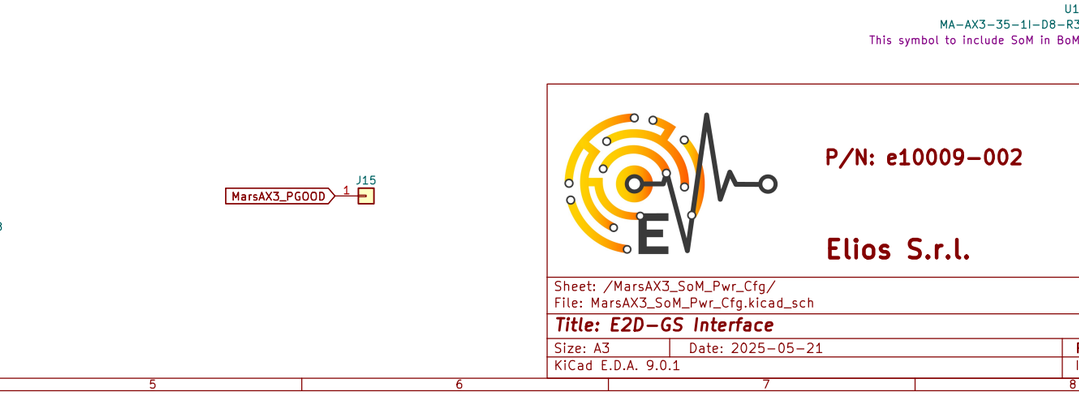
Sheet: /CMOS_power/
File: power.kicad_sch

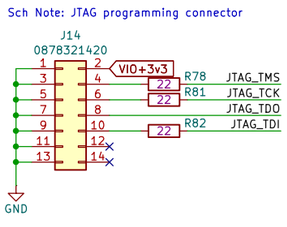
Title: E2D-GS Interface

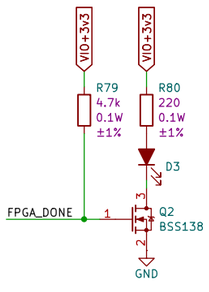
Size: A3	Date: 2025-05-21	Rev: 00A
KiCad E.D.A. 9.0.1		Id: 5/9











U17
MA-AX3-35-1I-D8-R3
This symbol to include SoM in BoM

P/N: e10009-002

Elios S.r.l.

Sheet: /MarsAX3_SoM_Pwr_Cfg/		
File: MarsAX3_SoM_Pwr_Cfg.kicad_sch		
Title: E2D-GS Interface		
Size: A3	Date: 2025-05-21	**Rev: 00A**
KiCad E.D.A. 9.0.1		Id: 6/9

Place capacitor near REF35102

Place each capacitor near each ADA4522

Current generator load
(inside ET2-GS) ~200 Ohm

J17 – J24 to check current value.
Amperometer between J17.1 and J21.2.
Normale operation J17.1 short with J17.2

Iout = VREF_1V024 / (R152 + RV2)
Iout from ~10uA to ~1.7uA

Unused

Unused

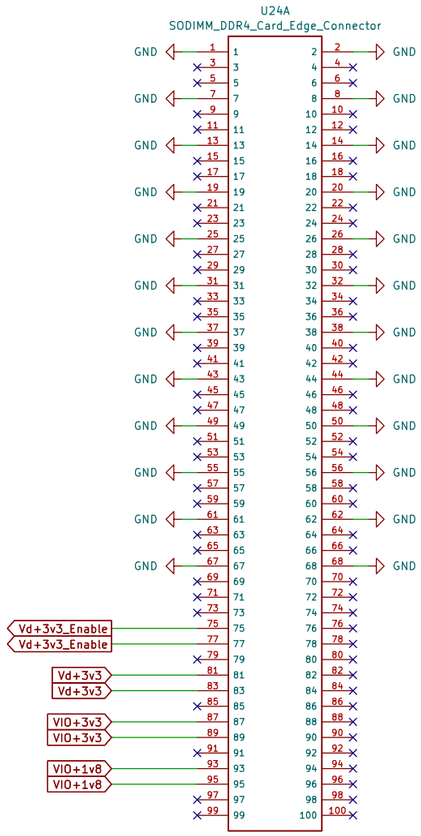
P/N: e10009-002

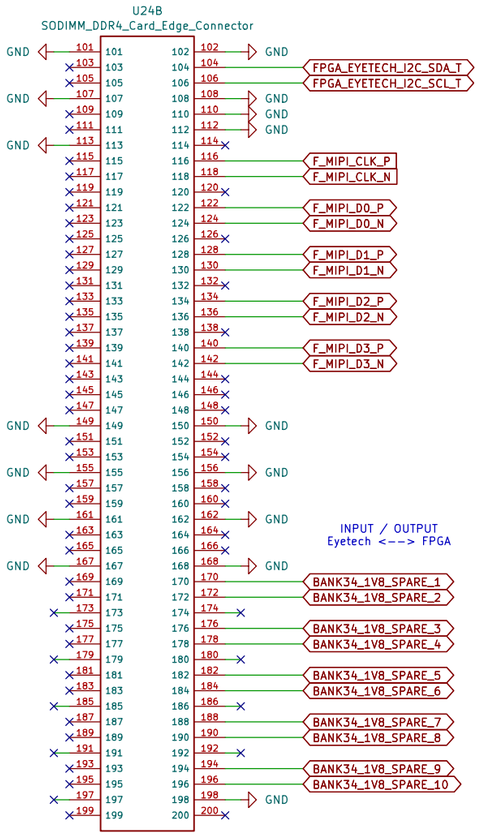
Elios S.r.l.

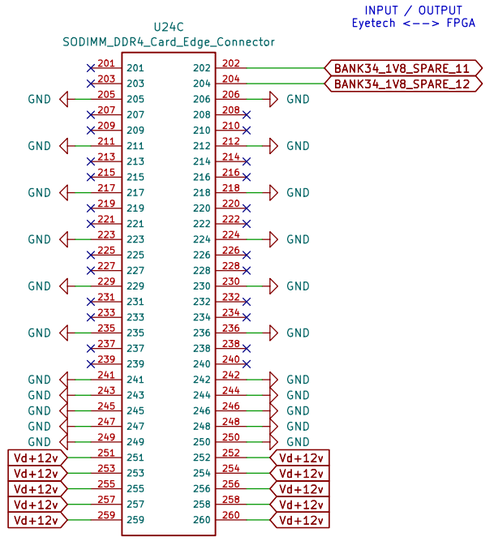
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File: current_generators.kicad_sch

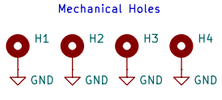
Title: E2D-GS Interface

Size: A3	Date: 2025-05-21	Rev: 00A
KiCad E.D.A. 9.0.1		Id: 7/9









P/N: e10009-002

Elios S.r.l.

Sheet: /DDR4_Edge_Connector/
File: DDR4_Edge_Connector.kicad_sch

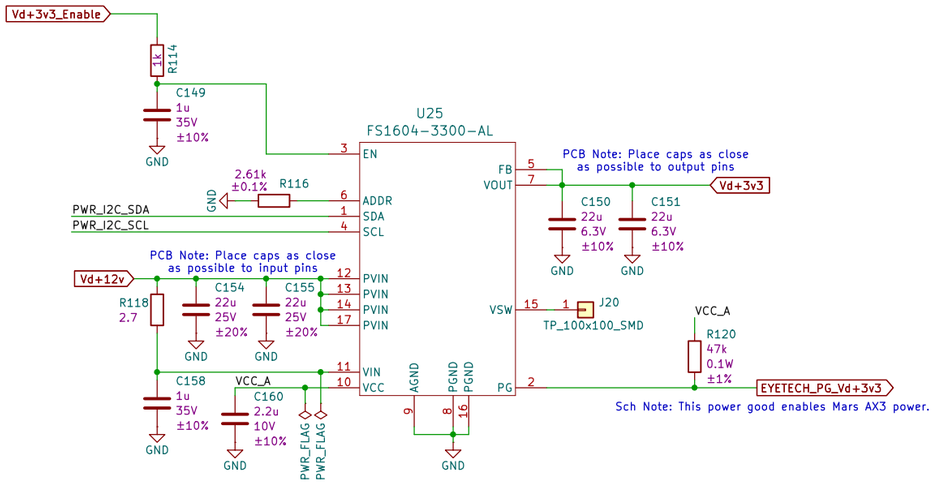
Title: E2D-GS Interface

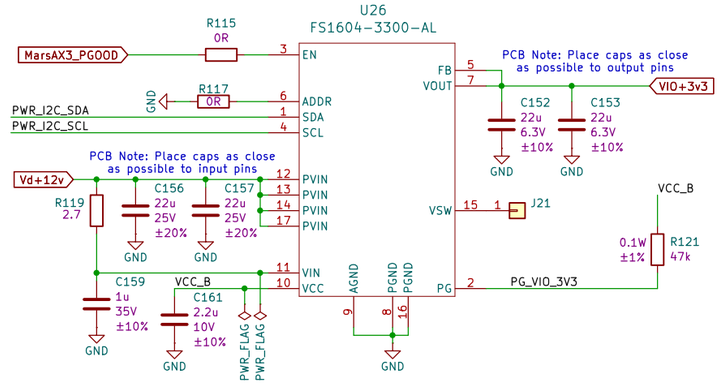
Size: A3	Date: 2025-05-21	**Rev: 00A**
KiCad E.D.A. 9.0.1		Id: 8/9

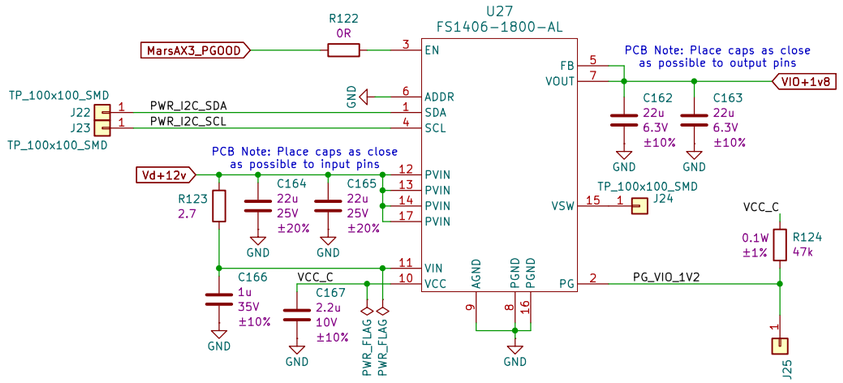
MarsAX3 SoM Power

MarsAX3 Bank 35 and 14 power supply

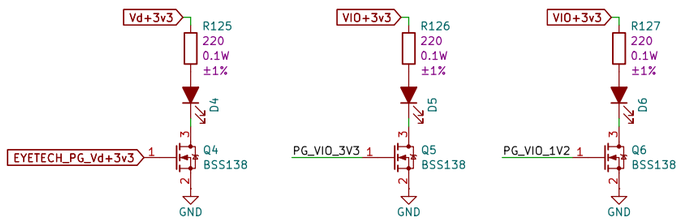
MarsAX3 Bank 34 power supply











P/N: e10009-002

Elios S.r.l.

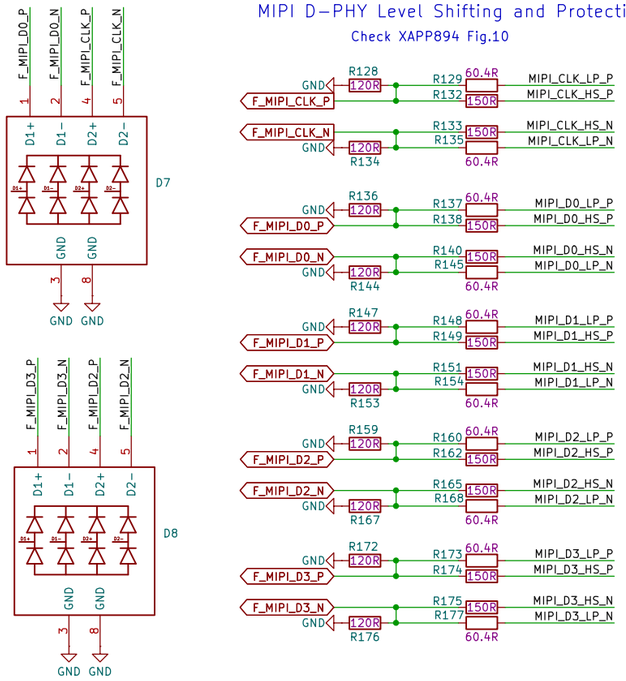
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Title: E2D-GS Interface

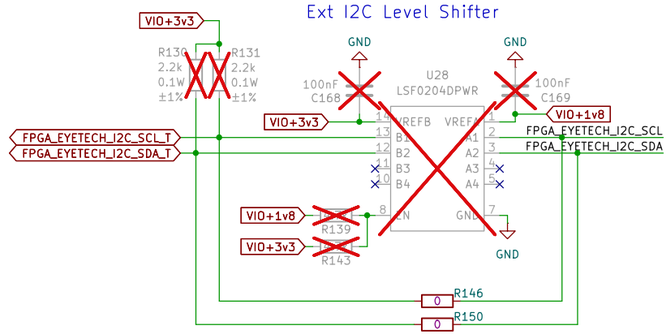
Size: A3	Date: 2025-05-21	**Rev: 00A**
KiCad E.D.A. 9.0.1		Id: 9/9

MIPI D-PHY Level Shifting and Protection

Check XAPP894 Fig.10



Ext I2C Level Shifter



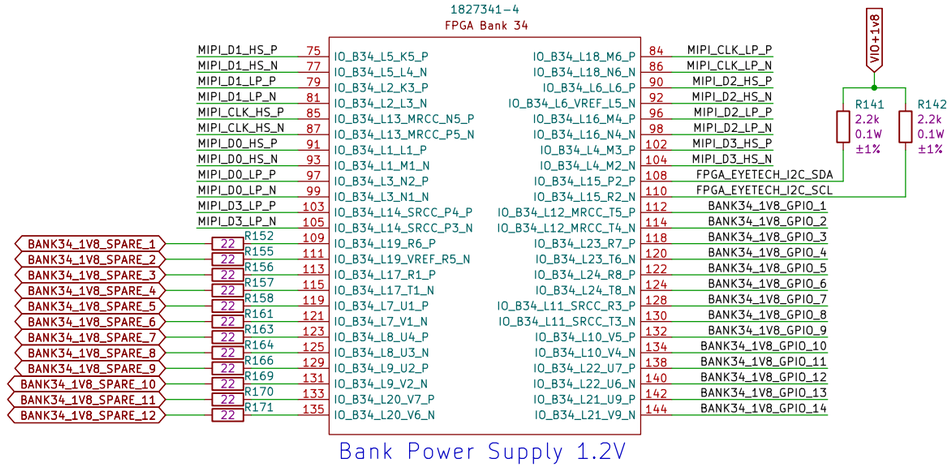
Debug & Spare I/O Connector

J26

TSW-108-07-G-D

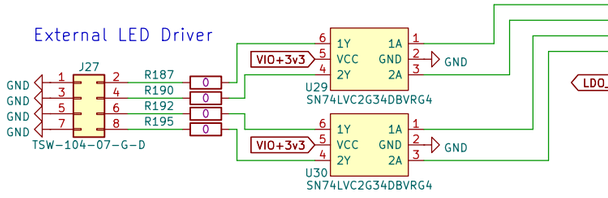
BANK34_1V8_GPIO_1	1	2	BANK34_1V8_GPIO_2
BANK34_1V8_GPIO_3	3	4	BANK34_1V8_GPIO_4
BANK34_1V8_GPIO_5	5	6	BANK34_1V8_GPIO_6
BANK34_1V8_GPIO_7	7	8	BANK34_1V8_GPIO_8
BANK34_1V8_GPIO_9	9	10	BANK34_1V8_GPIO_10
BANK34_1V8_GPIO_11	11	12	BANK34_1V8_GPIO_12
BANK34_1V8_GPIO_13	13	14	BANK34_1V8_GPIO_14
GND	15	16	VIO+1v8

U16B 1827341-4 FPGA Bank 34

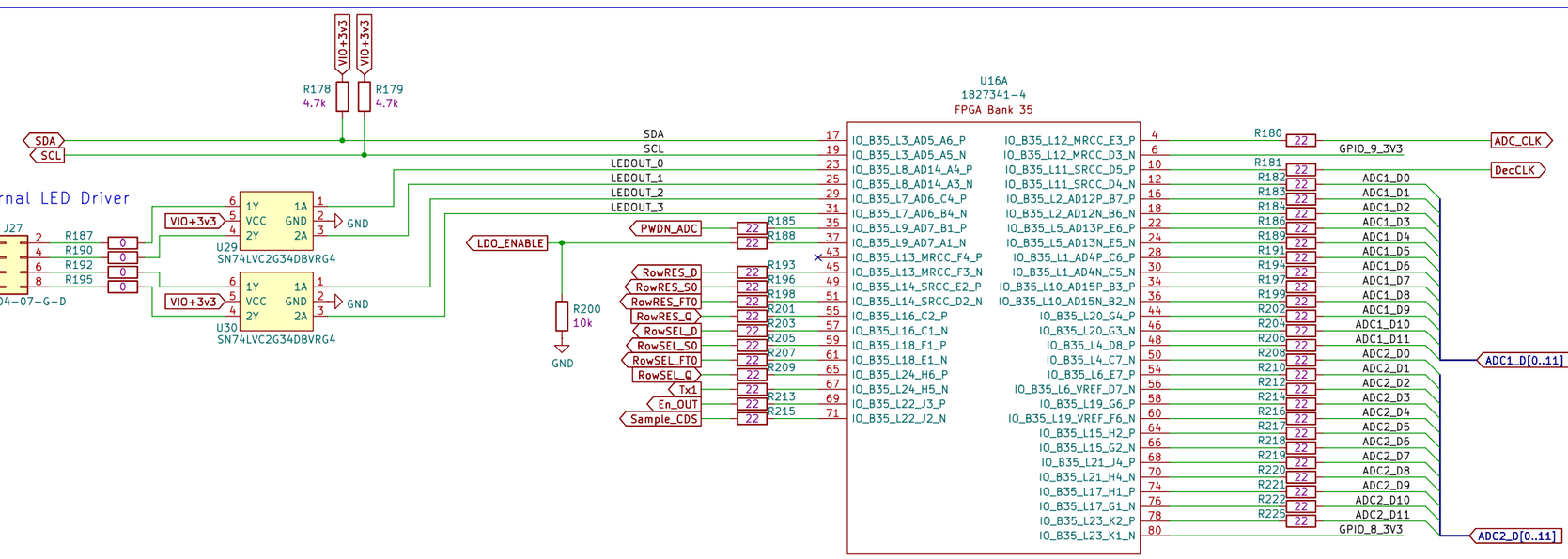


Bank Power Supply 1.2V

External LED Driver

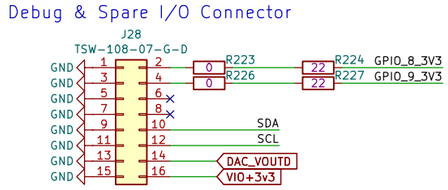


U16A 1827341-4 FPGA Bank 35

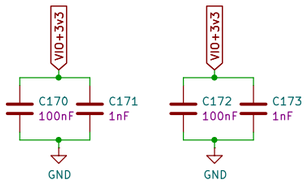


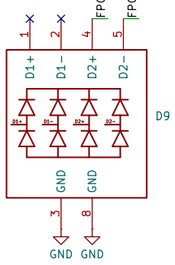
Bank Power Supply 3.3V

Debug & Spare I/O Connector

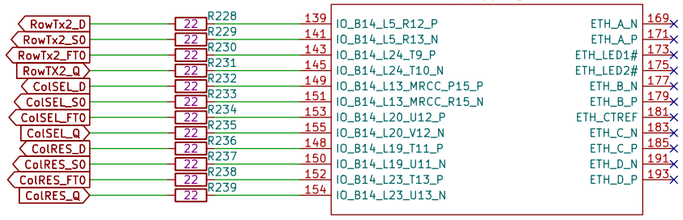


PCB Note: Place near SN74LVC2G34DBVRG4





U16E 1827341-4 FPGA Bank 14



Bank Power Supply 3.3V

P/N: e10009-002

Elios S.r.l.

Sheet: /MarsAX3_SoM_IO/
File: MarsAX3_SoM_IO.kicad_sch

Title: E2D-GS Interface

Size: A3	Date: 2025-05-21	**Rev: 00A**
KiCad E.D.A. 9.0.1		Id: 10/9